

Review-Guided Mini-Paper: Knowledge Unlearning for Mitigating Privacy Risks in Language Models

Source Paper

- Paper ID: openreview_sample_17
- Venue/source: ICLR_cc_2023_Conference
- arXiv: not available
- Original mean score: 0.55

Human Review Guidance Used

- The proposed method provides empirical privacy guarantees for large language models.
- The approach is straightforward and likely reproducible.

Review Evidence Excerpts

Review 1

{"clarity": null, "confidence": 0.5, "correctness": 0.6666666666666666, "ethics": null, "impact": null, "novelty": 1.0, "reproducibility": null, "review": {"limitations": null, "main_review": "main_review: While there may not be a lot of novelty in the methods designed in this work, I do believe that it has a significant contribution to the field of machine unlearning in large LMs. The quality of writing is good, and the authors provide code for reproducibility.", "paper_summary": "paper_summary: This work performs knowledge unlearning in large language models by attempting to fine-tune a converged model for a few epochs with a negative loss corresponding to the examples to the forgotten. As

Review 2

{"clarity": null, "confidence": 0.75, "correctness": 0.6666666666666666, "ethics": null, "impact": null, "novelty": 0.75, "reproducibility": null, "review": {"limitations": null, "main_review": "main_review: The paper is well-written and well-structured. The proposed method seems novel to me and the paper seems reproducible. However it is lacking justification for the choice of baselines and metrics.", "paper_summary": "paper_summary: This paper proposes a simple method for unlearning training samples, to comply with GDPR (and other privacy acts') right-to-be-forgotten statement, which gives each person the right to delete their data at any time they want. The proposed unlearning method for

Review 3

{"clarity": null, "confidence": 0.75, "correctness": 0.6666666666666666, "ethics": null, "impact": null, "novelty": 0.75, "reproducibility": null, "review": {"limitations": null, "main_review": "main_review: The work appears novel, the writing is clear and is easy to read, and the approach is simple and straightforward enough that reproducing the results is likely feasible.", "paper_summary": "paper_summary: The authors propose a simple approach for unlearning specific token sequences in large pretrained language models (LMs). To unlearn a specific sequence of tokens x, the proposed approach simply negates the original training objective of minimizing the negative log-likelihood of x; this p

Paper-Only Baseline Artifact

Core contribution: This paper proposes a novel method for knowledge unlearning in language models to mitigate privacy risks without requiring full retraining.

Experiment plan: We will evaluate the effectiveness of the unlearning method by measuring performance metrics on language tasks before and after applying the unlearning process.

In this paper, we propose a novel method for knowledge unlearning in language models to address privacy risks without necessitating full retraining. Our approach utilizes unlikelihood training to target specific token sequences, effectively forgetting them while preserving general language modeling performance. We plan to evaluate the method's effectiveness by measuring performance metrics on various language tasks before and after applying the unlearning process. However, we acknowledge that the method's effectiveness may vary depending on the size and nature of the data being unlearned, and potential degradation in performance requires thorough investigation. The claims made are specifically relevant to language models and may not extend to other types of machine learning models.

Human-Review-Guided Artifact

Core contribution: This work presents a simple yet effective approach for unlearning specific token sequences in large pretrained language models, providing empirical privacy guarantees.

Method sketch: We employ gradient ascent on the unlikelihood training objective to unlearn specific sequences, aiming for minimal degradation in performance.

Experiment plan: We will conduct experiments to assess the impact of unlearning on model performance, particularly focusing on the trade-off between privacy and accuracy.

Limitations: The justification for baseline choices and metrics needs to be strengthened to enhance the robustness of the findings.

Claim boundary: The findings are primarily relevant to large language models and may not generalize to other machine learning contexts.

In this work, we introduce a straightforward method for knowledge unlearning in large pretrained language models, aimed at mitigating privacy risks. Our approach employs gradient ascent on the unlikelihood training objective to effectively unlearn specific token sequences while striving for minimal degradation in performance. We plan to conduct experiments to evaluate the impact of unlearning on model performance, particularly focusing on the trade-off between privacy and accuracy. However, we recognize that the justification for our choice of baselines and metrics requires strengthening to enhance the robustness of our findings. The claims made are primarily relevant to large language models and may not extend to other machine learning contexts.

Short-Term Review Scores

gpt-4o-mini

- Winner: review_guided
- Context-control overall: 3
- Review-guided overall: 4
- Specific review value: The review-guided version provides empirical privacy guarantees and a clearer methodology, enhancing the overall quality.

claude-3-7-sonnet-latest

- Winner: tie
- Context-control overall: 3
- Review-guided overall: 3
- Specific review value: The review insights ('straightforward and reproducible', 'empirical privacy guarantees') are positive but shallow. The limitation about baseline justification is useful but generic. Neither version substantially outperforms the other in any dimension.

Later-Frontier Evidence

- Citation-frontier winner: paper_only

- Citation temporal pattern: short_term_positive_long_term_negative
- Semantic-frontier winner: review_guided_artifact
- Latent delayed-value candidate: False
- Frontier terms: unlearning, data, privacy, llms, machine, user, concern, parameter, knowledge, forget, forgetting, technique, dataset, request, existing, llm

Evidence Boundary

This PDF displays a concrete human-review-guided mini-paper artifact from the current pilot. It is not yet a full benchmark-executed deep reproduction.